In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

ALL INDIA TEST SERIES

FIITJEE

JEE(Main)-2017

ANSWERS, HINTS & SOLUTIONS

CONCEPT RECAPITULATION TEST – I
(Main)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**A**	**B**	**B**
2.	**A**	**C**	**D**
3.	**A**	**C**	**B**
4.	**A**	**D**	**C**
5.	**B**	**A**	**D**
6.	**A**	**B**	**A**
7.	**D**	**D**	**B**
8.	**C**	**C**	**C**
9.	**B**	**C**	**A**
10.	**A**	**B**	**A**
11.	**A**	**D**	**D**
12.	**B**	**A**	**B**
13.	**A**	**B**	**C**
14.	**C**	**C**	**C**
15.	**C**	**B**	**C**
16.	**D**	**B**	**A**
17.	**B**	**D**	**B**
18.	**D**	**B**	**A**
19.	**C**	**D**	**A**
20.	**C**	**B**	**D**
21.	**C**	**B**	**C**
22.	**C**	**C**	**A**
23.	**D**	**D**	**D**
24.	**A**	**C**	**A**
25.	**D**	**B**	**B**
26.	**D**	**D**	**D**
27.	**C**	**A**	**C**
28.	**B**	**B**	**A**
29.	**D**	**C**	**C**
30.	**B**	**B**	**C**

Physics

PART – I

SECTION – A

1. A

Sol. $\alpha x^2 + \beta x - t = 0 \Rightarrow x = \dfrac{-\beta \pm \sqrt{\beta^2 + 4\alpha t}}{2\alpha}$

$V = \dfrac{dx}{dt} = \dfrac{4\alpha}{2 \times 2\alpha \sqrt{\beta^2 + 4\alpha t}} = \pm \dfrac{1}{\sqrt{\beta^2 + 4\alpha t}}$

$a = \dfrac{dv}{dt} = \pm \dfrac{}{(\beta^2 + 4\alpha t)^{3/2}} \left(\dfrac{-1}{2}\right) \times 4\alpha = (-2\alpha)V^3$

2. A

Sol. Neutron law for cooling

Rate of decrease of temperature $\propto$ temperature difference between object and surrounding.

3. A

Sol. $1000 = \left(\dfrac{350}{350 - 50}\right) f$ …(i)

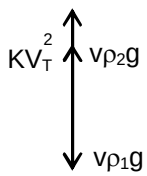
$f_1 = \left(\dfrac{350}{350 + 50}\right) f$ …(ii)

$f_1 = 750$ Hz

4. A

Sol. $v\rho_2 g + KV_T^2 = v\rho_1 g$

The ball is dynamic equilibrium.



5. B

Sol. $k' 2\ell = k 3\ell$

$k = \dfrac{3}{2} k$

6. A

Sol. $R = \dfrac{V^2}{g v_o} \cdot V = \dfrac{V^3}{g V_o}$; $R \propto V^3$

7. D

Sol. $\Delta Q = \Delta U + \Delta W$ …(i)

and

$\Delta U = nC_v(\Delta T)$ …(ii)

8. C

Sol. The centre of mass $= \frac{2R}{\pi}$

Linear velocity $= \left(\frac{2R}{\pi}\right)\left(\frac{V}{R}\right)$

$\because$ $V = r\omega$.

9. B

Sol: $Y = \frac{F.\ell}{A.\Delta\ell} \Rightarrow F = YA.\frac{\Delta\ell}{\ell}$

$= 2\times10^{11}\times\frac{1}{100}\times A$

$n = \frac{1}{2\times1.5}\sqrt{\frac{F}{Ad}} = \frac{1}{3}\sqrt{\frac{2\times10^9 A}{A\times7.7\times10^3}}$

$n = 170Hz$

10. A

Sol. $\frac{\nu}{4L_i} - \frac{\nu}{4L_2} = 5$ …(i)

$\frac{L_1}{L_2} = \frac{50}{51}$ …(ii)

11. A

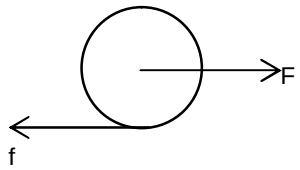
Sol. Escape velocity $= \sqrt{\frac{2GM}{R}}$

12. B

Sol. $B = 2\left[\frac{\mu_0 I}{4\pi r\cos45^0}\right]\left(\sin90^0 - \sin45^0\right) + \frac{\mu_0 I}{4\pi r}\left(\frac{\pi}{2}\right)$

13. A

Sol. $F = IBL$



$F - f = ma$

$fr = I\alpha$

$a = r\alpha$

14. C

Sol. $A > 2\theta_C$

15. C

Sol. $T = 2\pi\sqrt{\frac{\ell}{g_{eff}}}$; $g_{eff} = g + \frac{qE}{m}$ (with electric field)

16. D

Sol. For electron in electric field, magnitude of its momentum may increase, decrease or remain constant w.r.t. initial value.

17. B

Sol. $0.6 = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{20}{Q_1}$

$\Rightarrow$ $Q_1 = 50$, $W = Q_1 - Q_2 = 30$ J

18. D

Sol. I_L and I_C will be in opposite phases.

$I_{net} = I_L - I_C$

$= 0.6 - 0.4 = 0.2$ A

19. C

Sol. Variation of moment of inertia with temperature

$I = I_o (1 + 2\alpha \Delta T)$

C.O.A.M:

$I_i\omega_i = I_f \omega_f$

20. C

Sol. Polarization occurs for transverse waves and not for longitudinal waves.

21. C

Sol. $W = \frac{\lambda D}{d}$;

$\Delta W = \frac{\lambda \Delta D}{d}$

22. C

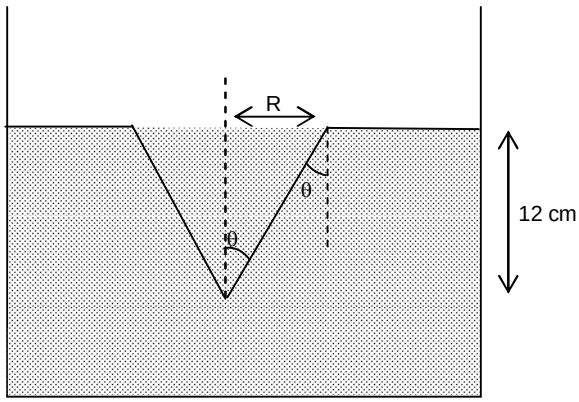
Sol. $V_{AC} = 2 \times \left(\frac{0.2}{1.8 + 0.2}\right)$ …(i)

Also, $V_{AC} = (1.5)\left(\frac{5}{15}\right)\left(\frac{\ell}{100}\right)$ …(ii)

23. D

Sol. $\theta = \theta_C$ (critical angle)

$\frac{R}{12} = \tan\theta_C$



24. A

Sol. $\varepsilon = B\ell v$ develops across rod PQ.

By Lenz's law, plate M get positively charged.

25. D

Sol. $A_1V_1 = A_2V_2$

$$\Rightarrow \; a\sqrt{2gh} = A\left(-\frac{dh}{dt}\right)$$

$$\Rightarrow \quad \frac{a}{A}\int_0^t dt = -A\int_{h_1}^{h_2}\frac{dh}{\sqrt{2gh}}$$

26. D

Sol. $$\frac{E_{4n}-E_{2n}}{E_{2n}-E_n} = \frac{\frac{E_1}{16n^2}-\frac{E_1}{4n^2}}{\frac{E_1}{4n^2}-\frac{E_1}{n^2}} = \frac{1}{4} = \text{constant}$$

27. C

Sol. $\Delta m = 0.026$ amu

$Q_{value} = 2.42$ MeV

$t = 1.28\times10^{12}$ s

28. B

Sol. $$\frac{hc}{\lambda} = hv = 2.5 + 1.7 = 4.2eV$$

$$v\lambda = C$$

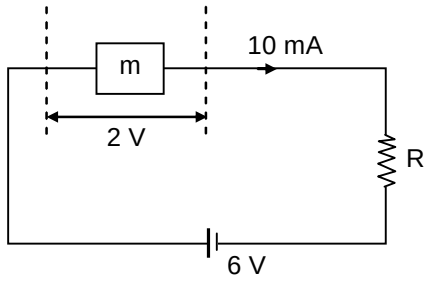
$$\& \; v'\frac{\lambda}{2} = C \Rightarrow v' = 2v$$

$$eV_0 = h(2v) - \varphi = 2\times4.2 - 2.5$$

$$V_0 = 5.9\,V$$

29. D

Sol. The potential drop across R is 4 V while current is 10 mA hence, $10 \times 10^{-3} \times R = 4$.
$R = 400\ \Omega$.



30. B

Sol. The electron density in doped semiconductor,

$$\eta_e = \frac{n_i^2}{n_h} = \frac{(3\times10^{16})^2}{4.5\times10^{22}} = 2 \times 10^{10}\ /m^3$$

Chemistry

PART – II

SECTION – A

1. B

Sol. Meq. of $FeSO_4$ = Meq. of $KMnO_4$
or, $N_1V_1 = N_2V_2$
or, $100 \times N_1 = 200 \times 0.1 \times 5$
$\therefore N_1 = 1$, also M = 1

2. C

Sol. It has unsymmetrical structures.

3. C

Sol. One mole of NH_3 occupy 22.4 L at NTP, not at 300 K and 1 atm.

4. D

Sol. Sulphur forms two sigma bonds with two lone pairs.

5. A

Sol. $BeSO_4$ forms complex with water i.e. $[Be(H_2O)_4]^{2+}$. Hence it is most soluble.

6. B

Sol. The unit of k will be $mol^{-1/2}L^{1/2}s^{-1}$.

7. D

Sol. K_a increases only by increasing temperature.

8. C

Sol. $K_C = \frac{[Y]}{[X]} = \frac{4}{6} = \frac{2}{3}$

9. C

Sol. For equilibrium system, $\Delta S(\text{system}) = -\Delta S(\text{surrounding})$

10. B

Sol. $(CH_3)_2 SiCl_2 \xrightarrow{H_2O} (CH_3)_2 Si(OH)_2 \xrightarrow{-H_2O} \text{Silicones}$

11. D

Sol. $K_P = \frac{p_y \times p_z}{p_x} = \frac{3 \times 3}{3} = 3$

$P_{equilibrium} = p_x + p_y + p_z = 3 + 3 + 3 = 9$ atm

12. A

Sol. $NaOH + CO \longrightarrow HCOONa$

13. B

Sol. $SO_3^{2-} + 2H^+ \longrightarrow SO_2 + H_2O$

14. C

Sol. HCOOH is stronger than CH_3COOH. Hence the solution of HCOOH will contain more no. of ions than that of CH_3COOH.

15. B

Sol. Zn^{2+} form colourless compounds in solution due to absence of d-d transition.

16. B

Sol. $\frac{\Lambda_m}{\Lambda_e} = 3 \Rightarrow \Lambda_m = 3\Lambda_e$

n-factor of $MgCl_2.KCl.6\ H_2O = 3$.

17. D

Sol. Due to weak Bi – H bond.

18. B

Sol. Halogen atoms except fluorine form compounds with valency 1, 3, 5 and 7.

19. D

Sol. Xe undergoes sp^3 hybridization and forms three sigma and three pi-bonds with a lone pair on it.

20. B

Sol. $NaOH \xrightarrow{CrO_2Cl_2} Na_2CrO_4 + H_2O + Cl^-$

$Pb^{2+} + 2Cl^- \longrightarrow PbCl_2 \downarrow$

$Pb^{2+} + CrO_4^{2-} \longrightarrow PbCrO_4 \downarrow$

21. B

Sol. One Na^+ is removed and remaining three(effective no. per unit cell) are present.

22. C

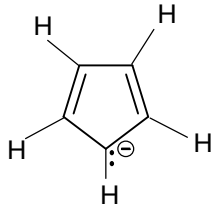
Sol. The fraction is $\frac{2}{3}$

23. D

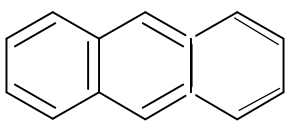
Sol. $E^0_{cell} = E^0_{cell} > 0$ (due to 1 M concentration of ions)

24. C

Sol. In cyclopentadienyl anion, each carbon is sp^2 hybridized



25. B
Sol. Fructose contains 3 chiral carbon atoms.

26. D
Sol.


27. A
Sol. Aliphatic amines are stronger base than aromatic amines

28. B
Sol. In xanthoproteic test, electrophylic nitration takes place.

29. C
Sol. $CH_3CHO \xrightarrow[KCN]{NH_4Cl} MeCH(NH_2)CN \xrightarrow[H_2O]{HCl} MeCH(NH_2)COOH$

30. B
Sol. The configuration of carbon atoms at 3, 4, 5 positions should be same.

Mathematics

PART – III

SECTION – A

1. B

Sol. $T_r = \cot^{-1}\left(3r^2 - \frac{5}{12}\right)$

$= \tan^{-1}\left(\frac{12}{36r^2 - 5}\right) = \tan^{-1}\left(\frac{12}{4 + 36r^2 - 9}\right)$

$= \tan^{-1}\left(\frac{3}{1 + \frac{9}{4}\left(4r^2 - 1\right)}\right)$

$= \tan^{-1}\left(\frac{\frac{3}{2}(2r+1) - \frac{3}{2}(2r-1)}{1 + \frac{3}{2}(2r+1).\frac{3}{2}(2r-1)}\right)$

$= \tan^{-1}\left(\frac{3}{2}(2r+1)\right) - \tan^{-1}\left(\frac{3}{2}(2r-1)\right)$

$\therefore S_n = \tan^{-1}\left(\frac{3}{2}(2n+1)\right) - \tan^{-1}\frac{3}{2}$

$= \tan^{-1}\left(\frac{12n}{18n+13}\right) = \cot^{-1}\left(\frac{18n+13}{12}\right)$

2. D

Sol. Equation of the curve is $y = x(1-x)$

3. B

Sol. $f(x) = \frac{\ln(\pi + x)}{\ln(e + x)}$

$f'(x) = \frac{\frac{\ln(e+x)}{(\pi+x)} - \frac{\ln(\pi+x)}{(e+x)}}{\left(\ln(e+x)\right)^2}$

Now, consider the function $g(y) = y \ln y$

$g'(y) = \ln y + 1$

$g'(y) > 0$ when $y > \frac{1}{e}$

$g(y)$ is an increasing function when $x > \frac{1}{e}$

$\Rightarrow (e+x)\ln(e+x) < (\pi+x)\ln(\pi+x) \qquad (x \geq 0)$

$\Rightarrow \dfrac{\ln(e+x)}{(\pi+x)} - \dfrac{\ln(\pi+x)}{(e+x)} < 0$

$\Rightarrow f'(x) < 0 \ \forall x \geq 0$

$\Rightarrow f(x)$ is decreasing is $[0, \infty)$

4. C

Sol. When $\cot x \geq 0$

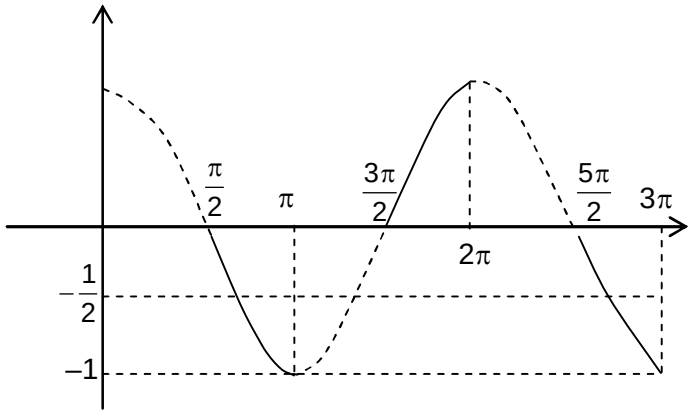
$\Rightarrow \dfrac{1}{\sin x} = 0$

$\Rightarrow$ No possible value of x.

When $\cot x < 0 \quad \Leftrightarrow \quad x \in \left(\dfrac{\pi}{2}, \pi\right) \cup \left(\dfrac{3\pi}{2}, 2\pi\right) \cup \left(\dfrac{5\pi}{2}, 3\pi\right)$

$\Rightarrow -\cot x = \cot x + \dfrac{1}{\sin x}$

$\Rightarrow \cos x = \dfrac{-1}{2}$



Number of solution = 2

5. D

Sol. $S = 2, S = 2016$

6. A

Sol. $x^3 + y^3 + (-1)^3 - 3(x)(y)(-1) = 0$

$\Rightarrow (x + y - 1) = 0$ OR $x^2 + y^2 - xy + x + y + 1 = 0$

7. B

Sol. $f(x)$ has minimum value at 2. $f(1)$ will be smaller than f (4) because 1 is closer to 2 as compared to 4.

8. C

Sol.

	M1	M2	M3	M4
$\frac{\sigma}{x}$	$\frac{2}{30}$	$\frac{3}{57}$	$\frac{4}{82}$	$\frac{2}{28}$

which ever is less is most consistent.

9. A

Sol. $\arg\left(\frac{z-2i}{z+2i}\right)=\frac{\pi}{6} \quad \Rightarrow \arg(z-2i)-\arg(z+2i)=\frac{\pi}{6}$

$\Rightarrow \tan^{-1}\left(\frac{y-2}{x}\right)-\tan^{-1}\left(\frac{y+2}{x}\right)=\frac{\pi}{6}$

$\Rightarrow \frac{xy-2x-xy-2x}{x^2+y^2-4}=\frac{1}{\sqrt{3}}$

$\Rightarrow x^2+y^2-4=-4\sqrt{3}\,x$

$\Rightarrow x^2+y^2+4\sqrt{3}\,x-4=0$

$\Rightarrow \left(x+2\sqrt{3}\right)^2+y^2=12+4=4^2$

$\therefore \text{centre}\left(-2\sqrt{3},0\right),$ radius = 4

10. A

Sol. $S=\frac{\sin\left(8^o-4^o\right)}{\sin 8^o \sin 4^o}+\frac{\sin\left(16^o-8^o\right)}{\sin 16^o \sin 8^o}+.....+\frac{\sin\left(8192^o-4096^o\right)}{\sin 8192^o \sin 4096^o}$

$\Rightarrow S=\cot 4^o-\cot 8192^o$

11. D

Sol. $\lim\limits_{h\to 0}\frac{f\left(h^3+3h+2\right)-f(2)}{f\left(2h-2h^2+1\right)-f(1)}=\lim\limits_{h\to 0}\frac{f'\left(h^3+3h+2\right).\left(3h^2+3\right)}{f'\left(2h-2h^2+1\right).(2-4h)}=\frac{6.3}{4.2}=\frac{9}{4}$

(Using L' Hospital Rule)

12. B

Sol. $I=\int\frac{\sin 3x}{32}dx$

13. C

Sol. $\left(x^2+4x+1\right)^2+a-1=0$

$\therefore \quad \left(x^2+4x+1\right)^2=1-a$

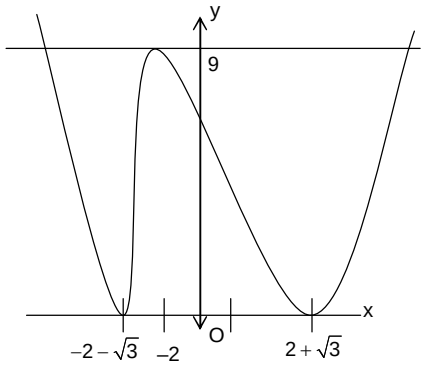
$f(x)=\left(x^2+4x+1\right)^2$

For four distinct solutions

$1-a\in(0,9)$

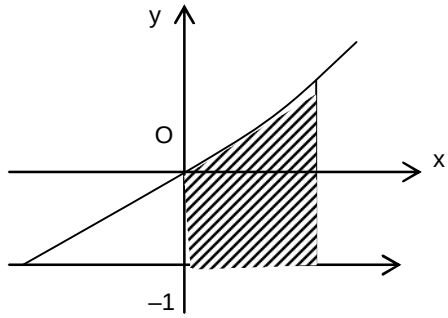
$-a\in(-1,8)$

$a\in(-8,1)$



14. C

Sol. $\dfrac{f''(x)}{f'(x)} = 1$



Integrating $\ln f'(x) = x + c,\ f'(0) = 1 \Rightarrow c = 0$

$f'(x) = e^x$

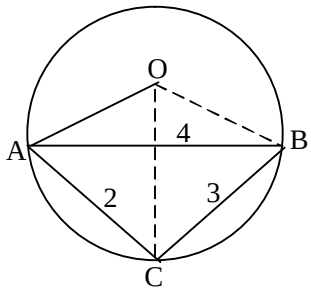
$f(x) = e^x + k,\ f(0) = 0 \Rightarrow k = -1$

$f(x) = e^x - 1$

Area $= \int_0^1 \left(e^x - 1 + 1\right) dx = e^x \Big]_0^1 = e - 1$

15. C

Sol. Using the property that equal chords subtends equal angles at centre of circle, then problem can be converted to the diagram in adjoining figure



$AB = 4, AC = 2, BC = 3$

$\angle ABC = \alpha/2$

$\cos(\alpha/2) = \dfrac{9+16-4}{2\times 3\times 4} = \dfrac{7}{8} \Rightarrow \cos\alpha = 2\cos^2(\alpha/2) - 1$

$= 2\times\dfrac{49}{64} - 1 = \dfrac{98-64}{64} \Rightarrow \cos\alpha = \dfrac{34}{64} = \dfrac{17}{32}$

16. A

Sol. $\cos(\theta_1 - \theta_2) + \cos(\theta_2 - \theta_3) + \cos(\theta_3 - \theta_1) \geq -\dfrac{3}{2}$

17. B

Sol. $\cos^{-1} x = t \Rightarrow x \in [-1, 1]$ and $t \in [0, \pi]$

$\log_{10}\sqrt{5t-1} + \dfrac{1}{2}\log_{10}(2t+3) + \dfrac{1}{2}\log_{10} 5 = 1;\ \left(t > \dfrac{1}{5} \text{and} t > -\dfrac{3}{2}\right)$

$\log_{10}\left((5t-1)(2t+3).5\right) = 2$

$(t-1)(10t+23) = 0 \Rightarrow t = 1$ or $t = -\dfrac{23}{10}$ (rejected)

$\cos^{-1} x = 1 \Rightarrow x = \cos 1$

18. A

Sol. $\dfrac{dy}{dx} = \dfrac{4}{2}e^{\frac{1-x}{2}} + 1 + x + x^2 = 2e^{\left(\frac{1-x}{2}\right)} + x^2 + x + 1$

$g'(y) = \dfrac{dx}{dy} = \dfrac{1}{2e^{-\left(\frac{x-1}{2}\right)} + x^2 + x + 1}$

When $y = -\dfrac{7}{6}$ then $x = 1$

$\dfrac{dy}{dx}\Big]_{x=-7/6} = \dfrac{1}{2+3} = \dfrac{1}{5}$

19. A

Sol. $= \dfrac{z\left(z\bar{z}-1\right)+z+\bar{z}}{z\bar{z}} = \left(z-\dfrac{1}{\bar{z}}\right)+\dfrac{1}{\bar{z}}+\dfrac{1}{z} = z+\dfrac{1}{z}$

20. D

Sol. Number of ways = (Number of one-one function from set A = $\{s_1, s_2 \ldots s_6\}$ to A such that $f(s_i) \neq s_i \times 6! = D_6 \times 6!$

21. C

Sol. The given plane passes through $\vec{a}$ and is parallel to the vector $\vec{b}-\vec{a}$ and $\vec{c}$. So it is normal to $\left(\vec{b}-\vec{a}\right)\times\vec{c}$. Hence, its equation is $\left(\vec{r}-\vec{a}\right).\left(\left(\vec{b}-\vec{a}\right)\times\vec{c}\right)=0$ or

$\vec{r}.\left(\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right)=\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]$

The length of the perpendicular from the origin to this plane is $\dfrac{\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]}{\left|\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right|}$

22. A

Sol. $A^n = \begin{bmatrix} \cos n\theta & \sin n\theta \\ -\sin n\theta & \cos n\theta \end{bmatrix}$

$\therefore \lim\limits_{n\to\infty} \dfrac{A^n}{n} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

23. D

Sol. We have $T = \int_0^{\ln 2} \dfrac{\left(3e^{3x}+2e^{2x}-e^x\right)-\left(e^{3x}+e^{2x}-e^x+1\right)}{e^{3x}+e^{2x}-e^x+1}dx$

$= \left[\ln\left(e^{3x}+e^{2x}-e^x+1\right)-x\right]_0^{\ln 2}$

$= \ln\dfrac{11}{2}-\ln 2 = \ln\dfrac{11}{4} \Rightarrow e^T = e^{\ln\frac{11}{4}} = \dfrac{11}{4}$

24. A

Sol. D.R's of PQ are x_1-1, y_1-1, z_1-1

$\Rightarrow 1(x_1-1)+1(y_1-1)+1(z_1-1)=0$

$\Rightarrow x_1+y_1+z_1=3$

Also $x_1+2y_1+3z_1=4$

$x_1+y_1+z_1=3$

$\Rightarrow (x_1, y_1, z_1)$ is a point on the two planes

$x+2y+3z=4$

$x+y+z=3$

$\therefore$ line of intersection of these two planes will be the required locus which is

$\dfrac{x}{1}=\dfrac{y-5}{-2}=\dfrac{z+2}{1}$

25. B

Sol. Parabola is of the form $(x+3)^2+\left(y-\frac{1}{2}\right)^2=\left(\frac{3x+4y-\frac{1}{2}}{5}\right)^2$ having focus $\left(-3,\frac{1}{2}\right)$ and point of intersection of lines given by the equation $2x^2-3y^2+xy-3x-7y+2=0$ can be obtained by solving equations $4x+y-3=0$ and $-6x+x-7=0$ (partial differentiation method), i.e $(1-1)\equiv$ centre (point of intersection of normals).

Hence required circle is $(x-1)^2+(y+1)^2=(1+3)^3+\left(-1-\frac{1}{2}\right)^2$

Or $x^2+y^2-2x+2y+2=16+\frac{9}{4}$ i.e. $4x^2+4y^2-8x+8y-65=0$

26. D

Sol. Asymptotes equations are $x^2-xy+k=0$

27. C

Sol. $\frac{dy}{dx}=\frac{y}{x}\left(\ell n\frac{y}{x}-1\right)$

Put $\frac{y}{x}=t$

28. A

Sol. Number of ways $=10\left(4^5-4\right)$

29. C

Sol. The expression can be interpreted as the square of distance between the points $(\tan A, \cot A)$ and $(\cos B, \sin B)$

The minimum value of this distance is the minimum distance between the curves $xy=1$ and $x^2+y^2=1$

30. C

Sol. For every selection of 6 numbers without loss of generality there will be 5 favorable cases out of every 20 cases.